COMMUNICATION-EFFICIENT SPARSE REGRESSION: A ONE-SHOT APPROACH

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We devise a one-shot approach to distributed sparse regression in the high-dimensional setting. The key idea is to average "debiased" lasso estimators. We show the approach converges at the same rate as the lasso when the dataset is not split across too many machines.

1. Introduction. Explosive growth in the size of modern datasets has fueled interest in distributed statistical learning. For examples, we refer to Boyd et al. (2011); Dekel et al. (2012); Duchi, Agarwal and Wainwright (2012); Zhang, Duchi and Wainwright (2013) and the references therein. The problem arises, for example, when working with datasets that are too large to fit on a single machine and must be distributed across multiple machines. The main computational bottleneck in the distributed setting is usually communication between machines/processors, so we must design algorithms that are communication-efficient. One-shot approaches that require only a *single* round of communication are highly desirable. However, the restriction of a single round of communication poses a significant challenge to algorithm design.

In the context of one-shot approaches, the simplest, and most popular, is *averaging*: each machine computes a local estimator $\hat{\theta}_l$ with the portion of the data stored locally, and a "master" averages the local estimators to produce an aggregate estimator: $\bar{\theta} = \frac{1}{k}\sum_{l=1}^{k}\hat{\theta}_l$. Averaging was first studied by Mcdonald et al. (2009) for multinomial regression. They derive non-asymptotic error bounds on the estimation error that show averaging reduces the variance of the local estimators, but has no effect on the bias (from the centralized solution). In follow up work, Zinkevich et al. (2010) studied a variant of averaging where each machine computes a local estimator with stochastic gradient descent (SGD) on a random subset of the dataset. They show, among other things, that their estimator converges to the centralized estimator.

More recently, Zhang, Duchi and Wainwright (2013) studied averaged empirical risk minimization (ERM). They show that the mean square error

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(MSE) of the averaged ERM decays like $O\big(\frac{1}{\sqrt{nk}}+\frac{k}{nk}\big)$, where k is the number of machines and nk is the total number of samples. Thus, so long as $k \lesssim \sqrt{nk}$, the averaged ERM matches the convergence rate of the centralized ERM. More recently, Rosenblatt and Nadler (2014) studied the optimality of averaged ERM in two asymptotic settings: $n \to \infty, p$ fixed and $p, n \to \infty, \frac{p}{n} \to \mu \in (0,1)$, where $n = nk/k$ is the number of samples per machine. They show that in the $n \to \infty, p$ fixed setting, the averaged ERM is first-order equivalent to the centralized ERM. However, when $p, n \to \infty$, the averaged ERM is suboptimal (versus the centralized ERM). Our work can be thought of as generalizing the averaged empirical risk minimization method to regularized empirical risk minimization.

Chen and Xie (2012) and Wang, Peng and Dunson (2014) also consider sparse linear regression in the distributed setting. They propose an averaging estimator that obtains the same convergence rate as the centralized lasso (lasso on all the data), but relies heavily on the irrepresentable condition and the beta-min condition (minimum signal strength condition). Our algorithm also obtains the same convergence rate as the centralized lasso, but does not rely on the aforementioned assumptions. Recall that the irrepresentable condition and beta-min condition together ensure variable selection consistency (Hastie, Tibshirani and Wainwright, 2015), so under these assumptions the lasso on local machine recovers the true support S. By computing the least squares solution restricted to S on each local machine, and then averaging, we obtain an estimator with the desired convergence rate. In contrast, our algorithm of averaging debiased estimators works even when the signal is weak and the predictors are correlated, which makes recovering the true support impossible. In concurrent work, Zhao, Kolar and Liu (2014) study inference in quantile regression, and propose a divide-and-conquer algorithm for hypothesis testing.

We devise an averaged estimator that converges at the same rate as the centralized lasso estimator. The key idea is to "debias" the local lasso estimators, followed by averaging and thresholding. Our averaged estimator achieves

1. $\big\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\big\|_\infty \lesssim \sqrt{\frac{\log p}{nk}}$,
2. $\big\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\big\|_2 \lesssim \sqrt{\frac{s\log p}{nk}}$,
3. $\big\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\big\|_1 \lesssim \sqrt{\frac{s^2\log p}{nk}}$.

where n is the number of points per machine, and k is the total number of machines. The aforementioned rates are minimax-optimal up to log factors and match or improve the convergence rate of the centralized lasso.

In section 2, we review the sparse linear model, the lasso estimator, and the debiased lasso estimator. Section 3 contains the main contributions of this paper. We present the *averaged debiased lasso* estimator, a one shot algorithm that obtains the same convergence rate as the centralized lasso. Section 5 shows the empirical performance of the averaged debiased lasso estimator.

1.1. *Notation:.* For a positive integer n, $[n]$ is the set $\{1,\ldots,n\}$. For functions $f(n)$ and $g(n)$, $f(n) \lesssim g(n)$ means $f(n) \leq cg(n)$ for some universal constant $c \in (0,\infty)$.

2. Background on lasso and debiased lasso.

2.1. *Sparse linear model.* Consider the linear model

$$y = X\beta^\star + \epsilon, \tag{2.1}$$

where $X \in \mathbf{R}^{n\times p}$ is a matrix of predictors, and $y \in \mathbf{R}^n$ are the responses. To keep things simple, we assume

1. the regression coefficients $\beta^\star \in \mathbf{R}^p$ are s-sparse, i.e. all but s components of $\beta^\star$ are zero.
2. the components of $\epsilon \in \mathbf{R}^n k$ are independent subgaussian random variables with mean zero and subgaussian norm σ_y.

2.2. *The lasso.* Given the predictors and responses, we seek a "good" estimate of the regression coefficients $\beta^\star$ in the ℓ_2 or ℓ_∞ norm. A common regularized M-estimator for sparse regression is the *lasso* by Tibshirani (1996) (also known as *basis pursuit denoising* by Chen, Donoho and Saunders (2001) in signal processing):

$$\hat\beta = \arg\min_\beta \frac{1}{2n}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1. \tag{2.2}$$

There is a well-developed theory of the lasso that says, under suitable assumptions on X, the lasso estimator $\hat\beta$ is nearly as close to $\beta^\star$ as the least squares solution given the support of $\beta^\star$. nkamely, when $n \gtrsim s\log p$, the error $\|\hat\beta - \beta^\star\|_2 \sim \sqrt{\frac{s\log p}{n}}$, and the bias and variance each make up a constant proportion of the mean-square error. Since the bias and variance are of the same order, we gain (almost) nothing by averaging the biased lasso estimators.[1] The key to overcoming the bias of the averaged lasso estimator is to "debias" the lasso estimators before averaging.

[1] We refer to Section 2.2 in Javanmard and Montanari (2013) for a more formal discussion of the bias of the lasso estimate.

2.3. *The debiased lasso.* The *debiased lasso estimator* is given by

$$\hat{\beta}^d = \hat{\beta} + \frac{1}{n} M X^T (y - X\hat{\beta}), \tag{2.3}$$

where $\hat{\beta}$ is the lasso estimator and $M \in \mathbf{R}^{p\times p}$ is an approximate inverse to $\hat{\Sigma} = \frac{1}{n} X^T X$. Intuitively, the local debiased lasso trades bias for variance. When $\hat{\Sigma}$ is invertible, setting $M = \hat{\Sigma}^{-1}$ gives the ordinary least squares (OLS) coefficients $(X^TX)^{-1}X^Ty$. Another way to interpret the debiased lasso is a corrected estimator that compensates for the bias incurred by incurred by shrinkage. By the optimality conditions of (2.2), the correction term $\frac{1}{n}X^T(y - X\hat{\beta})$ is a subgradient of $\lambda \|\cdot\|_1$ at $\hat{\beta}$. By adding a term proportional to the subgradient, the debiased lasso compensates for the bias incurred by ℓ_1 regularization. The debiased lasso has previously been used to perform inference on the regression coefficients in high-dimensional regression models. We refer to Javanmard and Montanari (2013); van de Geer et al. (2013); Zhang and Zhang (2014); Belloni, Chernozhukov and Hansen (2011) for details.

The correction term $\frac{1}{n}MX^T(y - X\hat{\beta})$ depends on M plays a key role in debiasing the lasso. In the high-dimensional setting, $\hat{\Sigma}$ is singular. Javanmard and Montanari (2013) suggest forming M by the solution of a convex optimization problem. The j-th row of M is the solution to

$$\begin{aligned} &\underset{m \in \mathbf{R}^p}{\text{minimize}} && m\hat{\Sigma}m^T \\ &\text{subject to} && \|\hat{\Sigma}m^T - e_j\|_\infty \le \tau. \end{aligned} \tag{2.4}$$

As we shall see, $m\hat{\Sigma}m^T$ is proportional to the variance of the debiased lasso, while $\|\hat{\Sigma}m^T - e_j\|_\infty$ is proportional to its bias. Thus (2.4) trades-off the bias and variance of the debiased estimator. Javanmard and Montanari (2013) refer to $\|\hat{\Sigma}m^T - e_j\|_\infty$ as the *generalized coherence* between X and M.

DEFINITION 2.1 (Generalized coherence). *Given $X \in \mathbf{R}^{nk\times p}$, let $\hat{\Sigma} = \frac{1}{n}X^TX$. The* generalized coherence *between $\hat{\Sigma}$ and $M \in \mathbf{R}^{p\times p}$ is*

$$\mathrm{GC}(\hat{\Sigma}, M) = \max_j \|\hat{\Sigma}m_j^T - e_j\|_\infty.$$

The parameter τ in (2.4) should large enough to keep the problem feasible, but as small as possible to keep the bias (of the debiased lasso) small. As we shall see, when the rows of X are independent subgaussian random vectors, setting $\tau \sim \sqrt{\frac{\log p}{nk}}$ is usually large enough to keep (2.4) feasible.

LEMMA 2.2 (Javanmard and Montanari (2013)). *When the rows of* $X \in \mathbf{R}^{n\times p}$ *are subgaussian with covariance* Σ, *subgaussian norm* σ_x, *and* $16\frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}\sigma_x^4 n > \log p$, *the event*

$$\mathcal{E}_{\mathrm{GC}}(\hat{\Sigma}) = \left\{\mathrm{GC}(\hat{\Sigma}, \Sigma^{-1}) \leq \tau\right\} \text{ for } \tau = \tfrac{8}{\sqrt{c_1}}\sqrt{\tfrac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}}\sigma_x^2\sqrt{\tfrac{\log p}{n}}, \tag{2.5}$$

occurs with probability at least $1 - 2p^{-2}$ *for some* $c_1 > 0$.

As we shall see, the bias of the debiased lasso estimate decays faster than its variance under suitable conditions on $\hat{\Sigma}$. Thus averaging debiased lasso estimates should improve the convergence rate. To make preceeding the statement rigorous, we require $\hat{\Sigma}$ to satisfy the *restricted eigenvalue (RE) condition* over $S := \mathrm{supp}(\beta^\star) \subset [\,p\,]$.

DEFINITION 2.3 (Restricted eigenvalue (RE) condition). *For any* $S \subset [\,p\,]$, *let*

$$C(S) = \{\Delta \in \mathbf{R}^p \mid \left\|\Delta_{S^c}\right\|_1 \leq 3\left\|\Delta x_S\right\|_1\}.$$

We say $\hat{\Sigma}$ *satisfies the RE condition over* $S \subset [\,p\,]$ *when*

$$\Delta^T\hat{\Sigma}\Delta \geq \mu\|\Delta_S\|_2^2 \text{ for any } \Delta \in C(S). \tag{2.6}$$

The RE condition is one of many conditions that require $\hat{\Sigma}$ to be positive definite over a restricted set of directions. A closely related condition, called the *compatibility condition*, replaces the right side of (2.6) with $\frac{\phi}{s}\|\Delta_S\|_1^2$, for some $\phi > 0$. We refer to Chapter 6 of Bühlmann and Van De Geer (2011) for details.

A natural question is whether there are many matrices that satisfy the condition. When the rows of X are *i.i.d.* Gaussian random vectors with mean zero and covariance Σ, Raskutti, Wainwright and Yu (2010) show there are constants $\mu_1, \mu_2 > 0$ such that

$$\tfrac{1}{n}\left\|X\Delta\right\|_2^2 \geq \mu_1\left\|\Delta\right\|_2^2 - \mu_2\tfrac{\log p}{n}\left\|\Delta\right\|_1^2 \text{ for any } \Delta \in \mathbf{R}^p$$

with probability at least $1 - c_2\exp\left(-c_2 n\right)$. Their result implies the RE condition for any $S \subset [\,p\,]$ with constant $\frac{\mu_1}{2}$ as long as $n > 32\frac{\mu_2}{\mu_1}\left|S\right|\log p$. Thus the RE condition holds with high probability for any design matrix with *i.i.d.* Gaussian rows, even when there are dependencies among the covariates. The result was extended to subgaussian designs by Rudelson and Zhou (2013), also allowing for dependencies among the covariates. We summarize their result in a lemma.

LEMMA 2.4. *When the rows of the design* $X \in \mathbf{R}^{n\times p}$ *are subgaussian with covariance* Σ*, subgaussian norm* σ_x*, and* $n > 4000 s' \sigma_x^2 \log(\frac{60\sqrt{2}ep}{s'})$*, where* $s' = (1 + 30000\frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)})s$*, and* $p > s'$*, the event*

$$\mathcal{E}_{\mathrm{RE}}(X) = \Big\{\Delta^T \hat{\Sigma}\Delta \geq \tfrac{1}{2}\lambda_{\min}(\Sigma)\|\Delta_S\|_2^2 \text{ for any } \Delta \in C(S)\Big\} \tag{2.7}$$

occurs with probability at least $1 - 2e^{-\frac{n}{4000\sigma_x^4}}$.

PROOF. The result follows from Rudelson and Zhou (2013) Theorem 6 by setting $\delta = \frac{1}{\sqrt{2}}, k_0 = 3$ and bounding $\max_j \|Ae_j\|_2^2$ and $K(s_0, k_0, \Sigma^{\frac{1}{2}})$ from above by $\lambda_{\max}(\Sigma)$ and $\frac{1}{\sqrt{\lambda_{\min}(\Sigma)}}$. □

Given the RE condition, the debiased lasso estimate is consistent for a suitable choice of the parameter λ. Intuitively, λ should be set large enough to dominate the "empirical process part" of the problem: $\frac{1}{n}\left\|X^T y\right\|_\infty$. As we shall see, setting $\lambda \sim \sigma_y\sqrt{\frac{\log p}{n}}$ is typically large enough.

LEMMA 2.5. *In the fixed design setting, when the components of* ϵ *are independent subgaussian random variables with mean zero and subgaussian norm* σ_y*, the event*

$$\mathcal{E}_\lambda(X, y) = \left\{\tfrac{1}{n}\|X^T\epsilon\|_\infty \leq \lambda\right\} \text{ for } \lambda = \max_j \hat{\Sigma}_{j,j}^{\frac{1}{2}}\sigma_y\sqrt{\tfrac{2\log p}{c_2 n}} \tag{2.8}$$

occurs with probability at least $1 - ep^{-2}$.

In practice, X is usually normalized so $\hat{\Sigma} = \frac{1}{n}X^T X$ has unit diagonal entries to put all the predictors on the same scale. When $\hat{\Sigma}$ satisfies the RE condition over $\mathrm{supp}(\beta^\star) \subset [p]$ and λ is large enough, the lasso and debiased lasso estimators are consistent.

LEMMA 2.6 (Negahban et al. (2012)). *When* $\hat{\Sigma}$ *satisfies the RE condition on* $\mathrm{supp}(\beta) \subset [p]$ *with constant* μ *and* $\mathcal{E}_\lambda(X, y)$ *occurs, the lasso satisfies*

$$\|\hat{\beta} - \beta\|_1 \leq \frac{3}{\mu}s\lambda \text{ and } \|\hat{\beta} - \beta\|_2 \leq \frac{3}{\mu}\sqrt{s}\lambda.$$

Given the convergence rate of the lasso estimator, we show that the bias of the debiased estimator is small.

LEMMA 2.7. *When $\hat{\Sigma}$ satisfies the RE condition on* $\operatorname{supp}(\beta) \subset [p]$ *with constant* μ, $(\hat{\Sigma}, M)$ *has generalized incoherence* τ, *and* $\mathcal{E}_\lambda(X, y)$ *occurs, the debiased lasso estimator is given by*

$$\hat{\beta}^d = \beta^\star + \frac{1}{n} M X^T \epsilon + \hat{\Delta}, \tag{2.9}$$

where $\|\hat{\Delta}\|_\infty \leq \frac{3\tau}{\mu} s\lambda$.

PROOF. We start by plugging the linear model into (2.3):

$$\hat{\beta}^d = \hat{\beta} + \frac{1}{n} M X^T (y - X\hat{\beta}) = \beta^\star + M\hat{\Sigma}(\beta^\star - \hat{\beta}) + \frac{1}{n} M X^T \epsilon.$$

By adding and subtracting $\hat{\Delta} = \beta^\star - \hat{\beta}$, we obtain

$$\hat{\beta}^d = \beta^\star + \frac{1}{n} M X^T (y - X\hat{\beta}) = \beta^\star + (M\hat{\Sigma} - I)(\beta^\star - \hat{\beta}) + \frac{1}{n} M X^T \epsilon.$$

We obtain (2.9) by setting $\hat{\Delta} = (M\hat{\Sigma} - I)(\beta^\star - \hat{\beta})$.

To show $\|\hat{\Delta}\|_\infty \leq \frac{3\tau}{\mu} s\lambda$, we apply Hölder's inequality to each component of $\hat{\Delta}$ to obtain

$$|(M\hat{\Sigma} - I)(\beta^\star - \hat{\beta})| \leq \max_j \|\hat{\Sigma} m_j^T - e_j\|_\infty \|\hat{\beta} - \beta^\star\|_1 \leq \tau \|\hat{\beta} - \beta^\star\|_1, \tag{2.10}$$

where τ is the generalized incoherence between X and M. By Lemma 2.6, when $\hat{\Sigma}$ satisfies the RE condition and $\mathcal{E}_\lambda(X, y)$ occurs, $\|\hat{\beta} - \beta^\star\|_1 \leq \frac{3}{\mu} s\lambda$. We combine the bound on $\|\hat{\beta} - \beta^\star\|_1$ with (2.10) to obtain the stated bound on $\|\hat{\Delta}\|_\infty$. □

We put the pieces together to deduce the bias of the debiased lasso estimator decays faster than the variance. In particular, setting λ and τ according to (2.8) and (2.7) implies $\|\hat{\Delta}\|_\infty$ decays like $O\big(\frac{s \log p}{n}\big)$. By comparison, $\frac{1}{n}\|M X^T \epsilon\|_\infty$ is the maxima of p subgaussian random variables with mean zero and variances $\frac{\sigma_y^2}{n} m_j \hat{\Sigma} m_j^T$, $j \in [p]$ which decays like $O\big(\sqrt{\frac{\log p}{n}}\big)$. Thus the bias term is of higher order than the variance term, when $n \gtrsim s^2 \log p$.

3. Averaging debiased lassos. In the distributed setting, we have nk pairs of covariates and responses $\{(x_i, y_i)\}_{i \in [nk]}$ distributed across k machines:

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_k \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix}.$$

The l-th machine has local predictors $X_l \in \mathbf{R}^{n_k \times p}$ and responses $y_l \in \mathbf{R}^{n_k}$. To keep things simple, we assume the data is evenly distributed across the machines, i.e. $n_1 = \cdots = n_k = \frac{nk}{k}$. The *averaged debiased lasso* (for lack of a better name) is given by averaging the local debiased lasso estimators:

$$\bar{\beta} = \frac{1}{k}\sum_{l=1}^{k} \hat{\beta}_l^d = \frac{1}{k}\sum_{l=1}^{k} \hat{\beta}_l + M_l X_l^T (y_l - X_l \hat{\beta}_l), \tag{3.1}$$

First, we study the error of the averaged debiased lasso in the ℓ_∞ norm.

LEMMA 3.1. *When $\hat{\Sigma}_1, \ldots, \hat{\Sigma}_l$ satisfy the RE condition on* $\operatorname{supp}(\beta^\star)$ *with constant μ, the pairs $\{(\hat{\Sigma}_l, M_l)\}_{l=1}^k$ have generalized incoherence $\alpha\sigma_x^2\sqrt{\frac{\log p}{n}}$, we set $\lambda_l = K_X^{\frac{1}{2}}\sigma_y\sqrt{\frac{3\log p}{c_2 n}}$, and $k \le p$, we have*

$$\|\tilde{\beta} - \beta^\star\|_\infty \le C\Big(K_M\sigma_y\sqrt{\frac{\log p}{nk}} + \alpha\frac{K_X^{\frac{1}{2}}}{\mu}\sigma_y\frac{s\log p}{n}\Big),$$

where C is some absolute constant, $K_M = \max_{j,l}\big(M_l\hat{\Sigma}_l M_l^T)\big)_{j,j}$, and $K_X = \max_{j,l}\hat{\Sigma}_{l;j,j}^{\frac{1}{2}}$, with probability at least $1 - ep^{-1}$.

PROOF. By Lemma 2.7,

$$\bar{\beta} - \beta^\star = \frac{1}{nk}\sum_{l=1}^{k} M_l X_l^T \epsilon_l + \frac{1}{k}\sum_{l=1}^{k}\hat{\Delta}_l.$$

We take norms to obtain

$$\|\bar{\beta} - \beta^\star\|_\infty \le \Big\|\frac{1}{nk}\sum_{l=1}^{k} M_l X_l^T \epsilon_l\Big\|_\infty + \frac{1}{k}\sum_{l=1}^{k}\|\hat{\Delta}_l\|_\infty.$$

Let

$$z_j = \tfrac{1}{nk}a_j^T\epsilon,\ a_j = \begin{bmatrix} X_1 M_1^T \\ \vdots \\ X_k M_l^T\end{bmatrix} e_j.$$

By Vershynin (2010), Proposition 5.10, $\mathbf{Pr}(|z_j| > t) \le e\exp\Big(-\frac{c_2 nk^2t^2}{\|a_j\|_2^2\sigma_y^2}\Big)$ for some absolute constant $c_2 > 0$. Further,

$$\|a_j\|_2^2 = \sum_{l=1}^{k}\big\|X_l M_l^T e_j\big\|_2^2 = n\sum_{l=1}^{k}\big(M_l\hat{\Sigma}_l M_l^T\big)_{j,j} \le K_M^2 nk,$$

where $K_M = \max_{j,l}\big(M_l\hat{\Sigma}_l M_l^T)\big)_{j,j}$. By a union bound over $j \in [\,p\,]$,

$$\mathbf{Pr}(\|z\|_\infty > t) \le e\exp\Big(-\tfrac{c_2 nkt^2}{K_M^2\sigma_y^2} + \log p\Big).$$

We set $t = K_M\sigma_y\sqrt{\frac{2\log p}{c_2 nk}}$ to deduce

$$\mathbf{Pr}\Big(\|z\|_\infty \ge K_M\sigma_y\sqrt{\tfrac{2\log p}{c_2 nk}}\,\Big) \le ep^{-1}.$$

By Lemma 2.7, the second term is bounded by $\frac{1}{k}\sum_{l=1}^k \frac{3\tau_l}{\mu_l}s\lambda$. By Lemma 2.5, when we set $\lambda_l = K_X^{\frac{1}{2}}\sigma_y\sqrt{\frac{3\log p}{c_2 n}}$, $l \in [\,k\,]$, the event $\cap_{l\in[k]}\mathcal{E}_{\lambda_k}(X_k, y_k)$ occurs with probability at least $1 - ekp^{-2} \ge 1 - ep^{-1}$. We put the pieces together to obtain

$$\|\bar{\beta} - \beta^\star\|_\infty \le K_M\sigma_y\sqrt{\frac{2\log p}{c_2 nk}} + \frac{3\sqrt{3}}{\sqrt{c_2}}\frac{K_X^{\frac{1}{2}}}{\mu}\alpha\sigma_y\frac{s\log p}{n}.$$

□

Lemma 3.1 hints at the performance of the averaged debiased lasso. In particular, we note the first term decays like $O\big(\sqrt{\frac{\log p}{nk}}\,\big)$, which matches the convergence rate obtained when estimating with all nk samples. When n is large enough, $\frac{s\log p}{n}$ is negligible compared to $\sqrt{\frac{\log p}{nk}}$, and the averaged debiased lasso converges at $O\big(\sqrt{\frac{\log p}{nk}}\,\big)$, which matches the l_∞ convergence rate of the centralized lasso.

Finally, we show the assumptions of Lemma 3.1 occur with high probability when the rows of X are independent subgaussian random vectors.

THEOREM 3.2. *When the rows of the design $X \in \mathbf{R}^{n\times p}$ are subgaussian with covariance Σ, subgaussian norm σ_x, and*

$$n > \max\big\{4000s'\sigma_x^2\log(\tfrac{60\sqrt{2}ep}{s'}), 8000\sigma_x^4\log p, \tfrac{3}{c_1}\max\{\sigma_x^2,\sigma_x\}\log p\big\},$$

where $s' = (1 + 30000\frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)})s$, $p > s'$, and $k < p$,

$$\|\bar{\beta} - \beta^\star\|_\infty \le C\Big(\max_j \Sigma_{j,j}^{-1}\sigma_y\sqrt{\frac{\log p}{nk}} + \frac{\sqrt{\kappa}\max_j\Sigma_{j,j}^{\frac{1}{2}}}{\lambda_{\min}(\Sigma)}\sigma_x^2\sigma_y\frac{s\log p}{n}\Big),$$

where $\kappa' = \frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}$, with probability at least $1 - 6p^{-1} - 2ke^{-\frac{n}{4000\sigma_x^4}}$.

PROOF. We start with the conclusion of Lemma 3.1:

$$\|\bar{\beta} - \beta^\star\|_\infty \le K_M \sigma_y \sqrt{\frac{2\log p}{c_2 nk}} + \frac{3\sqrt{3}}{\sqrt{c_2}} \frac{K_X^{\frac{1}{2}}}{\mu} \alpha \sigma_y \frac{s \log p}{n}$$

First, we show K_M and K_X are bounded with high probability.

LEMMA 3.3. *When the rows of $X \in \mathbf{R}^{n\times p}$ are independent subgaussian random vectors with mean zero, covariance Σ, subgaussian norm σ_x,*

$$\mathbf{Pr}\big(\max_j \Sigma_{j,\cdot}^{-1}\hat{\Sigma}\Sigma_{j,\cdot}^{-1} > 2\max_j \Sigma_{j,j}^{-1}\big) \le 2pe^{-c_1 \min\{\frac{n}{\sigma_x^2}, \frac{n}{\sigma_x}\}}$$

for some absolute constant $c_2 > 0$.

Since we minimize $\big(M_l \hat{\Sigma}_l M_l^T)\big)_{j,j}$ in (2.4), Lemma 3.3 (and union bound over $j \in [\,p\,]$) implies

$$\big(M_l \hat{\Sigma}_l M_l^T)\big)_{j,j} \le \max_j \big(\Sigma^{-1}\hat{\Sigma}_l\Sigma^{-1})\big)_{j,j} \le 2\max_j \Sigma_{j,j}^{-1} \text{ for each } k$$

with probability at least $1 - 2pe^{-c_1 \min\{\frac{n}{\sigma_x^2}, \frac{n}{\sigma_x}\}}$.

LEMMA 3.4. *When the rows of $X \in \mathbf{R}^{n\times p}$ are independent subgaussian random vectors with mean zero, covariance Σ, subgaussian norm σ_x,*

$$\mathbf{Pr}\big(\max_j \hat{\Sigma}_{j,j}^{\frac{1}{2}} > \max_j \sqrt{2}\Sigma_{j,j}^{\frac{1}{2}}\big) \le 2pe^{-c_1 \min\{\frac{n}{16\sigma_x^2}, \frac{n}{4\sigma_x}\}}$$

for some absolute constant $c_1 > 0$.

We put the pieces together to obtain the stated result:

1. By Lemma 3.3 (and a union bound over $l \in [\,k\,]$),

   $$\mathbf{Pr}\big(K_M \ge 2\max_j \Sigma_{j,j}^{-1}\big) \le 2kpe^{-c_1 \min\{\frac{n}{\sigma_x^2}, \frac{n}{\sigma_x}\}}.$$

   Since $k \le p$, when $n > \frac{3}{c_1}\max\{\sigma_x^2, \sigma_x\}\log p$, $\mathbf{Pr}\big(K_M \ge 2\max_j \Sigma_{j,j}^{-1}\big)$ at most $2p^{-1}$.
2. By Lemma 3.4 (and a union bound over $l \in [\,k\,]$),

   $$\mathbf{Pr}\big(K_X > 2\max_j \Sigma_{j,j}\big) \le 2kpe^{-c_1 \min\{\frac{n}{16\sigma_x^2}, \frac{n}{4\sigma_x}\}}$$

   When $n > \frac{3}{c_1}\max\{\sigma_x^2, \sigma_x\}\log p$, the right side is again at most $2p^{-1}$.

3. By Lemma 2.4, as long as

$$n > \max\{4000s'\sigma_x^2 \log(\tfrac{60\sqrt{2}ep}{s'}), 8000\sigma_x^4 \log p\},$$

$\hat{\Sigma}_1, \dots, \hat{\Sigma}_l$ satisfy the RE condition with probability at least $1 - 2ke^{-\frac{n}{4000\sigma_x^4}} \geq 1 - 2p^{-1}$.

4. By Lemma 2.2, $\mathbf{Pr}\big(\cap_{l\in[k]}\mathcal{E}_{\mathrm{GC}}(\hat{\Sigma}_l)\big) \geq 1 - 2p^{-2}$. Since $k < p$, the probability is at least $1 - 2p^{-1}$.

We plug in $K_M \leq 2\max_j \Sigma_{j,j}^{-1}$, $K_X \leq 2\max_j \Sigma_{j,j}$, $\alpha = \frac{8}{\sqrt{c_1}}\sqrt{\frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}}$, and $\mu = \frac{1}{2}\lambda_{\min}(\Sigma)$ to obtain

$$\|\bar{\beta} - \beta^\star\|_\infty \leq 2\max_j \Sigma_{j,j}^{-1}\sigma_y\sqrt{\frac{2\log p}{c_2 nk}} + \frac{48\sqrt{5}}{\sqrt{c_1c_2}}\frac{\sqrt{\kappa}\max_j \Sigma_{j,j}^{\frac{1}{2}}}{\lambda_{\min}(\Sigma)}\sigma_x^2\sigma_y\frac{s\log p}{n}.$$

□

3.1. *Sparsifying the averaged debiased lasso.* The averaged debiased lasso has one serious drawback versus the lasso: $\bar{\beta}$ is usually dense. The density of $\bar{\beta}$ detracts from the intrepretability of the coefficients and makes the estimation error large in the ℓ_2 and ℓ_1 norms. To remedy both problems, we propose to threshold the averaged debiased lasso:

$$\mathrm{HT}(\bar{\beta}, t) = \bar{\beta}_j \cdot \mathbf{1}_{\{|\bar{\beta}_j| \geq t\}}$$
$$\mathrm{ST}(\bar{\beta}, t) = \mathrm{sign}(\bar{\beta}_j) \cdot \max(|\bar{\beta}_j| - t, 0).$$

As we shall see, both hard and soft-thresholding give sparse aggregates that are close to $\beta^\star$ in ℓ_2 norm.

LEMMA 3.5. *As long as $t > \|\bar{\beta} - \beta^\star\|_\infty$, $\bar{\beta}^{\mathrm{ht}} = \mathrm{HT}(\bar{\beta}, t)$ satisfies*

1. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_\infty \leq 2t$,
2. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \leq 2\sqrt{2s}t$,
3. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_1 \leq 2\sqrt{2}st$.

The analogous result also holds for $\bar{\beta}^{\mathrm{st}} = \mathrm{ST}(\bar{\beta}, t)$.

PROOF. We have

$$\begin{aligned}\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_\infty &\leq \|\bar{\beta}^{\mathrm{ht}} - \bar{\beta}\|_\infty + \|\bar{\beta} - \beta^\star\|_\infty \\ &\leq t + \big\|\bar{\beta} - \beta^\star\big\|_\infty \\ &\leq 2t.\end{aligned}$$

Since $t > \left\|\bar{\beta} - \beta^\star\right\|_\infty$, $\bar{\beta}^{\mathrm{ht}}_j = 0$ whenever $\beta^\star_j = 0$. Thus $\bar{\beta}^{\mathrm{ht}}$ is s-sparse and $\beta^h - \beta^\star$ is $2s$-sparse. By the equivalence between the ℓ_∞ and ℓ_2/ℓ_1 norms,

$$\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \le 2\sqrt{2s}t$$
$$\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \le 2\sqrt{2s}t$$

The argument for $\bar{\beta}^{\mathrm{st}}$ is analogous. □

By combining the Lemma 3.5 with Theorem 3.2, we can show $\tilde{\beta}^{\mathrm{ht}}$ converges at the same rates as the centralized lasso.

THEOREM 3.6. *Under the conditions of Theorem 3.2, hard-thresholding at* $t \sim \max_j \Sigma^{-1}_{j,j}\sigma_y\sqrt{\frac{\log p}{nk}} + \frac{\sqrt{\kappa}\max_j \Sigma^{\frac{1}{2}}_{j,j}}{\lambda_{\min}(\Sigma)}\sigma_x^2\sigma_y\frac{s\log p}{n}$ *gives*

1. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_\infty \lesssim \max_j \Sigma^{-1}_{j,j}\sigma_y\sqrt{\frac{\log p}{nk}} + \frac{\sqrt{\kappa}\max_j \Sigma^{\frac{1}{2}}_{j,j}}{\lambda_{\min}(\Sigma)}\sigma_x^2\sigma_y\frac{s\log p}{n}$,
2. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \lesssim \max_j \Sigma^{-1}_{j,j}\sigma_y\sqrt{\frac{s\log p}{nk}} + \frac{\sqrt{\kappa}\max_j \Sigma^{\frac{1}{2}}_{j,j}}{\lambda_{\min}(\Sigma)}\sigma_x^2\sigma_y\frac{s^{\frac{3}{2}}\log p}{n}$,
3. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_1 \lesssim \max_j \Sigma^{-1}_{j,j}\sigma_y\sqrt{\frac{s^2\log p}{nk}} + \frac{\sqrt{\kappa}\max_j \Sigma^{\frac{1}{2}}_{j,j}}{\lambda_{\min}(\Sigma)}\sigma_x^2\sigma_y\frac{s^2\log p}{n}$.

PROOF. The convergence rates are straightforward consequences of Lemma 3.5 and Theorem 3.2. □

REMARK 3.7. *Theorem 3.6 shows that for* $k \lesssim \frac{n}{s^2\log p}$, *the term due to variance is dominant, so the convergence rates can be simplified to*

1. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_\infty \lesssim \sqrt{\frac{\log p}{nk}}$,
2. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \lesssim \sqrt{\frac{s\log p}{nk}}$,
3. $\|\bar{\beta}^{\mathrm{ht}} - \beta^\star\|_1 \lesssim \sqrt{\frac{s^2\log p}{nk}}$.

The corresponding rates for the centralized lasso estimator $\hat{\beta}$ *are*

1. $\|\hat{\beta} - \beta^\star\|_\infty \lesssim \sqrt{\frac{s\log p}{nk}}$,
2. $\|\hat{\beta} - \beta^\star\|_2 \lesssim \sqrt{\frac{s\log p}{nk}}$,
3. $\|\hat{\beta} - \beta^\star\|_1 \lesssim \sqrt{\frac{s^2\log p}{nk}}$.

The estimator $\bar{\beta}^{\mathrm{ht}}$ *improves on the lasso estimator in* ℓ_∞ *convergence rate, and match the lasso in* ℓ_2 *and* ℓ_1 *convergence rates. Furthermore,* $\bar{\beta}^{\mathrm{ht}}$ *can be computed in a communication-efficient manner via a one-shot averaging procedure.*

4. A distributed approach to debiasing. The averaged estimator we studied has the form

$$\bar{\beta} = \frac{1}{k}\sum_{l=1}^{k}\hat{\beta}_l + M_l X_l^T(y - X_l\hat{\beta}_l).$$

The estimator requires each machine to form M_l by the solution of (2.4). Since the dual of (2.4) is the ℓ_1-regularized quadratic program

$$\operatorname*{minimize}_{\eta\,\in\,\mathbf{R}^p} \frac{1}{2}\eta^T\hat{\Sigma}_l\eta - \hat{\Sigma}_l\eta + \tau|||\eta|||_1, \tag{4.1}$$

forming M_l is (roughly) p times as expensive as solving the local lasso problem. Forming $M_l, l \in [\,k\,]$ is the most expensive step (in terms of FLOPS) when evaluating the averaged estimator. To trim the cost of the debiasing step, we consider an estimator that forms a single M :

$$\tilde{\beta} = \frac{1}{k}\sum_{l=1}^{k}\hat{\beta}_l + \frac{1}{nk}M\sum_{l=1}^{k}X_l^T(y - X_l\hat{\beta}_l). \tag{4.2}$$

To evaluate (4.2), each machine communicates $\hat{\beta}_l$ and $\frac{1}{n}X_l^T(y - X_l\hat{\beta}_l)$ to a central server that forms $\frac{1}{k}\sum_{l=1}^{k}\hat{\beta}_l$ and $\frac{1}{nk}\sum_{l=k}^{k}X_l^T(y - X_l\hat{\beta}_l)$. The server sends the averages back to each machine, and each machine, given the averages, forms $\frac{p}{k}$ rows of M, m_j, and debiases $\frac{p}{k}$ components:

$$\tilde{\beta}_j = \frac{1}{k}\sum_{l=1}^{k}\hat{\beta}_j + m_j\bigg(\frac{1}{nk}\sum_{l=1}^{k}X_l^T(y - X_l\hat{\beta}_l)\bigg).$$

As we shall see, the rows of M only depends on data stored locally. Thus forming the estimator (4.2) requires two rounds of communication.

The question that remains is how to form M. We propose an estimator considered by van de Geer et al. (2013): nodewise regression on the design X. For some $j \in [\,p\,]$ that machine l is debiasing, the machine solves

$$\hat{\gamma}_j := \operatorname*{arg\,min}_{\gamma\in\mathbf{R}^{p-1}} \frac{1}{2n}\|x_{l,j} - X_{l,-j}\gamma\|_2^2 + \lambda_j\|\gamma\|_1,\ j \in [\,p\,],$$

where $X_{l,-j} \in \mathbf{R}^{n\times(p-1)}$ is X_l sans its j-th column $x_{l,j}$, and form

$$\hat{C} := \begin{bmatrix} 1 & -\hat{\gamma}_{1,2} & \dots & -\hat{\gamma}_{1,p} \\ -\hat{\gamma}_{2,1} & 1 & \dots & -\hat{\gamma}_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ -\hat{\gamma}_{p,1} & -\hat{\gamma}_{p,2} & \dots & -\hat{\gamma}_{p,p} \end{bmatrix},$$

where the components of $\hat{\gamma}_j$ are indexed by $k \in \{1, \dots, j-1, j+1, \dots, p\}$. Finally, we scale the rows of $\hat{C}$ by

$$\hat{T} := \mathbf{diag}(\big[\hat{\tau}_1, \dots, \hat{\tau}_p\big]), \hat{\tau}_j = \Big(\frac{1}{n}\|x_j - X_{-j}\hat{\gamma}_j\|_2^2 + \lambda_j\|\hat{\gamma}_j\|_1\Big)^{\frac{1}{2}}$$

to form $M = \hat{T}^{-2}\hat{C}$. Each row of M is given by

$$m_j = -\frac{1}{\hat{\tau}_j^2}\begin{bmatrix}\hat{\gamma}_{j,1} & \dots & \hat{\gamma}_{j,j-1} & 1 & \hat{\gamma}_{j,j+1} & \dots & \hat{\gamma}_{j,p}\end{bmatrix}.$$

Since $\hat{\gamma}_j$ and $\hat{\tau}_j$ only depend on X_l, they can be formed without communication.

van de Geer et al. (2013) show that when the rows of X are subgaussian and the precision matrix Σ^{-1} is sparse, m_j converges to $\Sigma^{-1}_{j,\cdot}$ at the usual convergence rate of the lasso. For completeness, we restate their result.

Lemma 4.1 (van de Geer et al. (2013), Theorem 2.4). *When the rows of X are subgaussian with covariance Σ and Σ^{-1} is sparse, i.e.* $\max_j\big(\frac{s_j\log p}{n}\big) \sim o(1)$, *where s_j is the sparsity of $\Sigma^{-1}_{j,\cdot}$, then for $\lambda_j \sim \sqrt{\frac{\log p}{n}}$,*

$$\|m_j - \Sigma^{-1}_{j,\cdot}\|_1 \lesssim s_j\sqrt{\tfrac{\log p}{n}}.$$

We show the estimator (4.2) also matches the convergence rate of the centralized lasso. The argument is similar to the proof of Theorem 3.2.

Theorem 4.2. *When the rows of X are subgaussian with covariance Σ and Σ^{-1} is sparse, i.e. $s_j \sim o\big(\frac{n}{\log p}\big)$, where s_j is the sparsity of $\Sigma^{-1}_{j,\cdot}$, and*

$$n > \max\big\{4000 s'\sigma_x^2\log(\tfrac{60\sqrt{2}ep}{s'}), 8000\sigma_x^4\log p, \tfrac{3}{c_1}\max\{\sigma_x^2, \sigma_x\}\log p\big\},$$

where $s' = (1 + 30000\frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)})s$, $p > s'$, and $k < p$, for $\lambda, \lambda_l \sim \sqrt{\frac{\log p}{n}}, l \in [\,k\,]$

$$\|\bar{\beta} - \beta^\star\|_\infty \sim O_P\bigg(\sqrt{\frac{\log p}{nk}} + \frac{\max_j s_j \log p}{n}\bigg).$$

Proof. We start by plugging in the linear model (2.1) into (4.2):

$$\begin{aligned}\tilde{\beta} &= \frac{1}{k}\sum_{l=1}^k \hat{\beta}_{l,j} - M\hat{\Sigma}_l(\hat{\beta}_l - \beta^\star) + \frac{1}{n}X_l^T\epsilon_l \\ &= \frac{1}{k}\sum_{l=1}^k \hat{\beta}_{l,j} - M\hat{\Sigma}_l(\hat{\beta}_l - \beta^\star) + \frac{1}{nk}X^T\epsilon.\end{aligned}$$

Subtracting $\beta^\star$ and taking norms, we obtain

$$\|\tilde{\beta} - \beta^\star\|_\infty \le \frac{1}{k}\sum_{l=1}^{k} \|(I - M\hat{\Sigma}_l)(\hat{\beta}_l - \beta^\star)\|_\infty + \Big\|\frac{1}{nk}X^T\epsilon\Big\|_\infty. \tag{4.3}$$

By an argument similar to the proof of Lemma 3.1, we have

$$\mathbf{Pr}\Big(\big\|\tfrac{1}{nk}X^T\epsilon\big\|_\infty \ge \sigma_y\sqrt{\tfrac{2\log p}{c_2 nk}}\Big) \le ep^{-1}.$$

We turn our attention to the first term in (4.3). It's straightforward to see each term in the sum is bounded by

$$\begin{aligned}
&\|(I - M\hat{\Sigma}_l)(\hat{\beta}_l - \beta^\star)\|_\infty \\
&\quad \le \|(I - \Sigma^{-1}\hat{\Sigma}_l)(\hat{\beta}_l - \beta^\star)\|_\infty + \|(\Sigma^{-1} - M)\hat{\Sigma}_l(\hat{\beta}_l - \beta^\star)\|_\infty \\
&\quad \le \max_j \|e_j^T - \Sigma_{j,\cdot}^{-1}\hat{\Sigma}_l\|_\infty \|\hat{\beta}_l - \beta^\star)\|_1 + \|(\Sigma_{j,\cdot}^{-1} - m_j)\|_1 \|\hat{\Sigma}_l(\hat{\beta}_l - \beta^\star)\|_\infty.
\end{aligned}$$

We put the pieces together to deduce each terms is $O\big(\frac{\max_j s_j \log p}{n}\big)$:

1. By Lemma 2.2, $\max_j \|e_j^T - \Sigma_{j,\cdot}^{-1}\hat{\Sigma}_l\|_\infty \lesssim \sqrt{\frac{\log p}{n}}$ with probability at least $1 - ep^{-2}$.
2. By Lemmas 2.4, 2.5, 2.6, $\|\hat{\beta}_l - \beta^\star\|_1 \lesssim \sqrt{s}\lambda$ with probability at least $1 - ep^{-2} - 2e^{-\frac{n}{4000\sigma_x^4}} \le 1 - (2+e)p^{-1}$.
3. By Lemma 4.1, $\|(\Sigma_{j,\cdot}^{-1} - m_j)\|_1 \lesssim O_P\Big(s_j\sqrt{\frac{\log p}{n}}\Big)$.
4. It's straightforward to see that

$$\|\hat{\Sigma}_l(\hat{\beta}_l - \beta^\star)\|_\infty \le \Big\|\frac{1}{n}X_l^T(y_l - X_l\hat{\beta}_l)\Big\|_\infty + \Big\|\frac{1}{n}X_l^T\epsilon_l\Big\|_\infty \lesssim \sqrt{\frac{\log p}{n}}.$$

By a union bound over $l \in [\,k\,]$, we obtain

$$\|\bar{\beta} - \beta^\star\|_\infty \sim O_P\bigg(\sqrt{\frac{\log p}{nk}} + \frac{\max_j s_j \log p}{n}\bigg).$$

□

By combining the Lemma 3.5 with Theorem 4.2, we can show that $\tilde{\beta}^{\mathrm{ht}} := \mathrm{HT}(\tilde{\beta}, t)$ for an appropriate t converges to $\beta^\star$ at the same rates as the centralized lasso.

THEOREM 4.3. *Under the conditions of Theorem 4.2, hard-thresholding at* $t \sim \sqrt{\frac{\log p}{n}} + \frac{\max_j s_j \log p}{n}$ *gives*

1. $\|\tilde{\beta}^{\mathrm{ht}} - \beta^\star\|_\infty \lesssim \sqrt{\frac{\log p}{n}} + \frac{\max_j s_j \log p}{n}$,
2. $\|\tilde{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \lesssim \sqrt{\frac{s \log p}{n}} + \frac{\sqrt{s}\max_j s_j \log p}{n}$,
3. $\|\tilde{\beta}^{\mathrm{ht}} - \beta^\star\|_1 \lesssim s\sqrt{\frac{\log p}{n}} + \frac{s\max_j s_j \log p}{n}$.

PROOF. The convergence rates are straightforward consequences of Lemma 3.5 and Theorem 4.2. □

REMARK 4.4. *Theorem 4.3 shows that for* $k \lesssim \frac{n}{s^2 \log p}$, *the term due to variance is dominant, so the convergence rates can be simplified to*

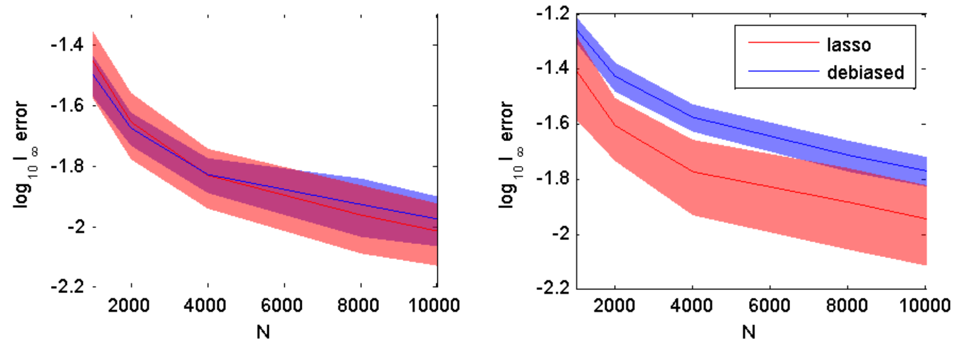
1. $\|\tilde{\beta}^{\mathrm{ht}} - \beta^\star\|_\infty \lesssim \sqrt{\frac{\log p}{nk}}$,
2. $\|\tilde{\beta}^{\mathrm{ht}} - \beta^\star\|_2 \lesssim \sqrt{\frac{s \log p}{nk}}$,
3. $\|\tilde{\beta}^{\mathrm{ht}} - \beta^\star\|_1 \lesssim \sqrt{\frac{s^2 \log p}{nk}}$.

Thus estimator $\tilde{\beta}^{\mathrm{ht}}$ *shares the advantages of* $\bar{\beta}^{\mathrm{ht}}$ *over the centralized lasso (cf. Remark 3.7). It also achieves computational gains over* $\bar{\beta}^{\mathrm{ht}}$ *by amortizing the cost of debiasing across* k *machines.*

5. Simulations. We validate our theoretical results with simulations. First, we study the estimation error of the averaged debiased lasso in ℓ_∞ norm. To focus on the effect of averaging, we grow the number of machines k linearly with the (total) sample size nk. In other words, we fix the sample size per machine n and grow the total sample size by adding machines. Figure 1 compares the estimation error (in ℓ_∞ norm) of the averaged debiased lasso estimator with that of the centralized lasso. We see the estimation error of the averaged debiased lasso estimator is comparable to that of the centralized lasso.

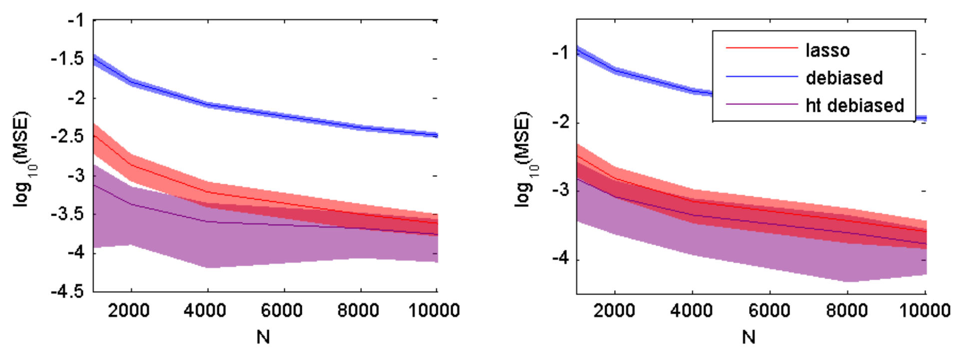
In the second set of simulations, we study the effect of thresholding on the estimation error in ℓ_2 norm. Figure 2 compares the estimation error incurred by the averaged estimator with and sans thresholding versus that of the centralized lasso. Since the averaged estimator is usually dense, its estimation error (in ℓ_2 norm) is large compared to that of the centralized lasso. However, after thresholding, the averaged estimator performs comparably versus the centralized lasso.

Finally, we study the effect of the number of machines on the estimation effor of the averaged estimator. To focus on the effect of the number of machines k, we fix the (total) sample size nk and vary the number of machines the samples are distributed across. Figure 3 shows how the estimation error (in ℓ_∞ norm) of the averaged estimator grows as the number of machines



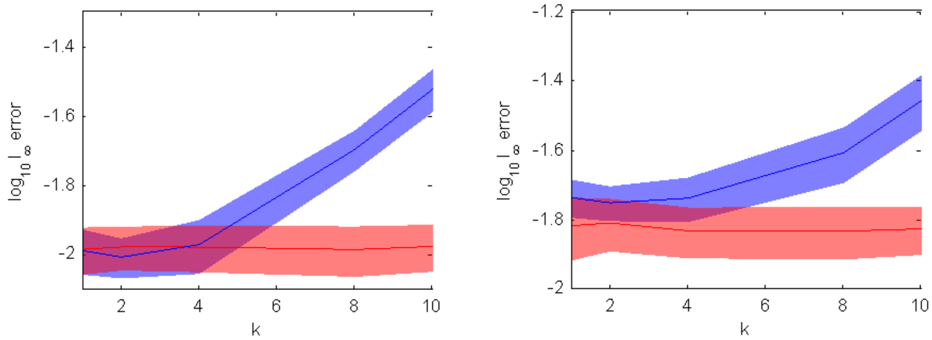
(a) $\Sigma = I$ (b) $(\Sigma)_{ij} = 0.5^{|i-j|}$

Fig 1: The estimation error (in ℓ_∞ norm) of the averaged debiased lasso estimator versus that of the centralized lasso when the predictors are Gaussian. In both settings, the estimation error of the averaged debiased estimator is comparable to that of the centralized lasso.



(a) $\Sigma = I$ (b) $(\Sigma)_{ij} = 0.5^{|i-j|}$

Fig 2: The estimation error (in ℓ_2 norm) of the averaged estimator with and sans thresholding versus that of the centralized lasso when the predictors are Gaussian. In both settings, thresholding reduces the estimation error by order(s) of magnitude. Although the estimation error of the averaged estimator is large compared to that of the centralized lasso, the thresholded averaged estimator performs comparably versus the centralized lasso.



(a) $\Sigma = I$ (b) $(\Sigma)_{ij} = 0.5^{|i-j|}$

Fig 3: The estimation error (in ℓ_∞ norm) of the averaged estimator as the number of machines k vary. When the number of machines is small, the error is comparable to that of the centralized lasso. However, when the number of machines exceeds a certain threshold, the bias term (which grows linearly in k) is dominant, and the performance of the averaged estimator degrades.

grows. When the number of machines is small, the estimation error of the averaged estimator is comparable to that of the centralized lasso. However, when the number of machines exceeds a certain threshold, the estimation error grows with the number of machines. This is consistent with the prediction of Theorem 3.2: when the number of machines exceeds a certain threshold, the bias term $\sim k\frac{s\log p}{nk}$ becomes dominant.

6. Summary and discussion. We devised an communication-efficient approach to distributed sparse regression in the high-dimensional setting. The key idea is first "debiasing" local lasso estimators, and then averaging the debiased estimators. We show that as long as the data is not split across too many machines, the averaged estimator achieves the convergence rate of the centralized lasso estimator. Further, the idea generalizes to most structured likelihood estimation problems.

In recent years, there has a been a flurry of work on establishing communication lower bounds for mean estimation in the Gaussian distribution. In other words, they establish the minimum communication C needed to obtain ℓ_2^2 risk R , where $\|\hat{\beta} - \beta^\star\|_2^2 \le R$ (Duchi et al., 2014; Garg, Ma and Nguyen, 2014). These results are not directly applicable to sparse linear regression, since they do not impose sparsity on the mean. In Ma (2015), the authors established that to obtain risk $R \le \frac{s\log p}{nk}$ at least $C \ge \frac{pk}{\log p}$, for some $C > 0$, bits of communication is needed. Our algorithm communicates $\tilde{O}(pk)$ bits to achieve risk of $\frac{s \log p}{nk}$, so is communication-optimal. Among algorithms that achieve the same estimation error, it uses the least amount of communication. Ma (2015) further established that $RC \geq \frac{ps}{n}$. We can achieve optimal risk/communication tradeoffs for values other than $C \sim pk$ by only averaging over C/p machines, instead of all k.

APPENDIX A: PROOFS OF AUXILIARY LEMMAS

Proof of Lemma 2.2. Let $z_i = \Sigma^{-\frac{1}{2}} x_i$. Then the generalized coherence between X and Σ^{-1} is given by

$$|||\Sigma^{-1}\hat{\Sigma} - I|||_\infty = |||\tfrac{1}{n}\textstyle\sum_{i=1}^n (\Sigma^{-\frac{1}{2}} z_i)(\Sigma^{\frac{1}{2}} z_i)^T - I|||_\infty.$$

Each entry of $\frac{1}{n}\sum_{i=1}^n (\Sigma^{-\frac{1}{2}} z_i)(\Sigma^{\frac{1}{2}} z_i)^T - I$ is a sum of independent subexponential random variables. Their subexponential norms are bounded by

$$\|(\Sigma^{-\frac{1}{2}} z_i)_j(\Sigma^{\frac{1}{2}} z_i)_k - \delta_{j,k}\|_{\psi_1} \leq 2\|(\Sigma^{-\frac{1}{2}} z_i)_j(\Sigma^{\frac{1}{2}} z_i)_k\|_{\psi_1}.$$

For any two subgaussian random variables X, Y, we have

$$\|XY\|_{\psi_1} \leq 2\,\|X\|_{\psi_2}\,\|Y\|_{\psi_2}\,.$$

Thus

$$\begin{aligned}\|(\Sigma^{-\frac{1}{2}} z_i)_j(\Sigma^{\frac{1}{2}} z_i)_k - \delta_{j,k}\|_{\psi_1} &\leq 4\|(\Sigma^{-\frac{1}{2}} z_i)_j\|_{\psi_2}\|(\Sigma^{\frac{1}{2}} z_i)_k\|_{\psi_2} \\ &\leq 4\sqrt{\tfrac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}}\sigma_x^2,\end{aligned}$$

where $\sigma_x = \|z_i\|_{\psi_2}$. By a Bernstein-type inequality for subexponential random variables,

$$\mathbf{Pr}(\tfrac{1}{n}\textstyle\sum_{i=1}^n (\Sigma^{-\frac{1}{2}} z_i)_j(\Sigma^{\frac{1}{2}} z_i)_k - \delta_{j,k} \geq t) \leq 2\exp(-c_1 \min\{\frac{nt^2}{s_x^4}, \frac{nt}{s_x^2}\}),$$

where $c_1 > 3$ is a constant and $s_x^2 = 4\sqrt{\frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)}}\sigma_x^2$. By setting $t = \frac{2s_x^2}{\sqrt{c_1}}\sqrt{\frac{\log p}{n}}$ (and since $s_x^4 n > \log p$),

$$\mathbf{Pr}(\tfrac{1}{n}\textstyle\sum_{i=1}^n (\Sigma^{-\frac{1}{2}} z_i)_j(\Sigma^{\frac{1}{2}} z_i)_k - \delta_{j,k} \geq \frac{2s_x^2}{\sqrt{c_1}}\sqrt{\frac{\log p}{n}}) \leq 2p^{-4}.$$

We obtain the desired bound by taking a union bound over the p^2 entries of $\frac{1}{n}\sum_{i=1}^n (\Sigma^{-\frac{1}{2}} z_i)(\Sigma^{\frac{1}{2}} z_i)^T - I$. □

Proof of Lemma 2.5. By Vershynin (2010), Proposition 5.10,

$$\mathbf{Pr}\left(\tfrac{1}{n}|x_j^T\epsilon| > t\right) \le e\exp\big(-\tfrac{c_2 n^2 t^2}{\sigma_y^2\|x_j^T\|_2^2}\big) \le e\exp\big(-\tfrac{c_2 n^2 t^2}{\sigma_y^2 \max_j \hat{\Sigma}_{j,j}}\big).$$

We the union bound over the p components of $\frac{1}{n}X^T\epsilon$ to obtain

$$\mathbf{Pr}\left(\tfrac{1}{n}\|X^T\epsilon\|_\infty > t\right) \le e\exp\big(-\tfrac{c_2 n^2 t^2}{\sigma_y^2 \max_j \hat{\Sigma}_{j,j}} + \log p\big).$$

We set $\lambda = \max_j \hat{\Sigma}_{j,j}^{\frac{1}{2}}\sigma_y\sqrt{\frac{3\log p}{c_2 n}}$ to obtain the desired conclusion. □

Proof of Lemma 3.3. We express

$$\Sigma_{j,\cdot}^{-1}\hat{\Sigma}\Sigma_{j,\cdot}^{-1} = \Sigma_{j,\cdot}^{-1}\hat{\Sigma}\Sigma_{j,\cdot}^{-1} - \Sigma_{j,j}^{-1} + \Sigma_{j,j}^{-1} = \frac{1}{n}\sum_{i=1}^n (x_i^T\Sigma_{\cdot,j})^2 - \Sigma_{j,j}^{-1} + \Sigma_{j,j}^{-1}.$$

Since the subgaussian norm of $z_i = \Sigma^{-\frac{1}{2}}x_i$ is σ_x, $x_i^T\Sigma_{\cdot,j}$ is also subgaussian with subgaussian norm bounded by

$$\|x_i^T\Sigma_{\cdot,j}\|_{\psi_2} \le \|\Sigma^{\frac{1}{2}}z_i\|_{\psi_2}\|\Sigma_{\cdot,j}\|_2 \le \sigma_x\sqrt{\Sigma_{j,j}^{-1}}.$$

We recognize $\frac{1}{n}\sum_{i=1}^n (x_i^T\Sigma_{\cdot,j})^2 - \Sigma_{j,j}^{-1}$ as a sum of *i.i.d.* subexponential random variables with subexponential norm bounded by

$$\|(x_i^T\Sigma_{\cdot,j})^2 - \Sigma_{j,j}^{-1}\|_{\psi_1} \le 2\|(x_i^T\Sigma_{\cdot,j})^2\|_{\psi_1} \le 4\|x_i^T\Sigma_{\cdot,j}\|_{\psi_2}^2 \le 4\sigma_x^2\Sigma_{j,j}^{-1}.$$

By Vershynin (2010), Proposition 5.16, we have

$$\mathbf{Pr}\big(\tfrac{1}{n}\textstyle\sum_{i=1}^n (x_i^T\Sigma_{\cdot,j})^2 - \Sigma_{j,j}^{-1} > t\big) \le 2\exp\big(-c_1\min\{\tfrac{nt^2}{16\sigma_x^2\left(\Sigma_{j,j}^{-1}\right)^2}, \tfrac{nt}{4\sigma_x\Sigma_{j,j}^{-1}}\}\big)$$

for some absolute constant $c_1 > 0$. For $t = \Sigma_{j,j}^{-1}$, the bound simplifies to

$$\mathbf{Pr}\big(\tfrac{1}{n}\textstyle\sum_{i=1}^n (x_i^T\Sigma_{\cdot,j})^2 - \Sigma_{j,j}^{-1} > \Sigma_{j,j}^{-1}\big) \le 2\exp\big(-c_1\min\{\tfrac{n}{16\sigma_x^2}, \tfrac{n}{4\sigma_x}\}\big).$$

We take a union bound over $j \in [p]$ to obtain the stated result. □

Proof of Lemma 3.4. We follow a similar argument as the proof of Lemma 3.3:

$$\hat{\Sigma}_{k;j,j} = \hat{\Sigma}_{j,j} = \hat{\Sigma}_{j,j} - \Sigma_{j,j} + \Sigma_{j,j} = \tfrac{1}{n}\textstyle\sum_{i=1}^n x_{i,j}^2 - \Sigma_{j,j} + \Sigma_{j,j}.$$

Since the $z_i = \Sigma^{-\frac{1}{2}} x_i$ is subgaussian with subgaussian norm σ_x, $x_{i,j}$ is also subgaussian with subgaussian norm bounded by

$$\|x_{i,j}\|_{\psi_2} \le \|\Sigma_{j,\cdot}^{\frac{1}{2}} z_i\|_{\psi_2} \le \sigma_x \sqrt{\Sigma_{j,j}}.$$

We recognize $\hat{\Sigma}_{j,j} - \Sigma_{j,j} = \frac{1}{n}\sum_{i=1}^n x_{i,j}^2 - \Sigma_{j,j}$ as a sum of $i.i.d.$ subexponential random variables with subexponential norm bounded by

$$\|\hat{\Sigma}_{j,j} - \Sigma_{j,j}\|_{\psi_1} \le 2\|x_{i,j}^2\|_{\psi_1} \le 4\|x_{i,j}\|_{\psi_2}^2 \le 4\sigma_x^2 \Sigma_{j,j}.$$

By Vershynin (2010), Proposition 5.16, we have

$$\mathbf{Pr}(\hat{\Sigma}_{j,j} - \Sigma_{j,j} > t) \le 2\exp\big(-c_1 \min\{\tfrac{nt^2}{16\sigma_x^2\Sigma_{j,j}^2}, \tfrac{nt}{\sigma_x\Sigma_{j,j}}\}\big)$$

for some absolute constant $c_1 > 0$. For $t = \Sigma_{j,j}$, the bound simplifies to

$$\mathbf{Pr}(\hat{\Sigma}_{j,j} - \Sigma_{j,j} > \Sigma_{j,j}) \le 2\exp\big(-c_1 \min\{\tfrac{n}{16\sigma_x^2}, \tfrac{n}{4\sigma_x}\}\big).$$

We take a union bound over $j \in [\,p\,]$ to obtain the stated result. □

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